

MoreOverleaf: A Mixed Reality Solution for Collaborative Document Workflows

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Figure 1: A novel Mixed Reality system to support improved collaborative writing experience

Abstract

Collaborative writing tools are often limited to 2D visualization of collaborator edits and comments. We present a novel Mixed Reality (MR) system that enhances collaborative document workflows by extending annotations and author interactions into 3D space. Our prototype introduces three immersive features: 3D comment boxes that spatially anchor comments around the user's monitor, a revision player visualizing the document history, and avatar-based representations of remote collaborators that convey presence and authorship. This approach aims to improve clarity, awareness, and engagement in collaborative writing.

CCS Concepts

• Human-centered computing → Mixed / augmented reality.

Keywords

Mixed Reality, Collaborative Writing, Annotation, Avatar

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1 Introduction

Collaborative writing has become increasingly common in both academia and industry, often supported by cloud-based editors like Google Docs or Overleaf. Despite their convenience, these tools present challenges in managing comments, resolving conflicting edits, and maintaining awareness of coauthors' activities across asynchronous workflows [Cho et al. 2025; Frassl et al. 2018].

Mixed Reality (MR) offers new possibilities for improving collaborative writing by augmenting the user's text editor with virtual content. Prior work has used MR to provide extended document views and awareness cues for coauthors [Li et al. 2019]. In addition, recent studies suggest that combining MR with physical displays can be more effective for document editing compared to working in fully virtual environments [Pavanatto et al. 2021]. However, efficiently visualizing complex revision histories and authorship context in a spatial and intuitive manner—while also enhancing clarity and social presence—remains an open challenge.

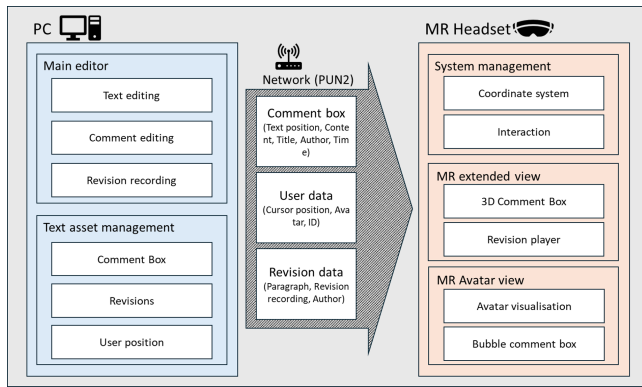


Figure 2: System overview

MoreOverleaf is an MR system that augments collaborative document workflows through three immersive features: spatially anchored 3D comment boxes, revision playback in space and time, and embodied avatars representing remote collaborators. By placing these virtual cues around the user's real monitor, we aim to improve editing visibility, social presence, and overall clarity during collaborative editing sessions.

2 Immersive MR Cue Implementation

Figure 2 shows an overview of the whole system. The setup includes a PC and a MagicLeap 2 see-through head-mounted display (HMD): the PC hosts the text editor, and the HMD displays immersive features. The software architecture comprises: (1) a text editor module on the PC for standard editing; (2) a text asset manager to store and organize data for MR cues; and (3) a network module that exchanges comments, revisions, and spatial data between the PC and HMD in real time. On the HMD side, a system manager oversees the MR environment with two visualization modules: the MR extended view and the MR avatar view.

2.1 MR Extended View

Our prototype supports collaborative editing through an immersive 3D comment box and revision player. The 3D comment box expands traditional 2D comment dialogs into the MR space (Figure 1, 3D immersive dialogue), enabling users to view and arrange comments beyond the physical screen. In the PC editor, users write comments linked to specific text areas, generating 2D comment boxes. These are associated with location and content data, which are transmitted in real time to the MR HMD. The headset displays the virtual boxes around the monitor, visually linking them to the corresponding text via 3D lines by converting screen coordinates into MR world space. Users can arrange the boxes using their hands or controller, supported by OpenXR interaction, freely positioning them in 3D space.

The revision player visualizes document history (Figure 1, Revision player) using a virtual timeline slider. Selecting a paragraph and adjusting the timeline slider replays its revision history spatially in MR. As the slider moves forward, comment boxes appear in the order they were added, supporting low-effort review of changes.

2.2 MR Avatar View

The MR avatar view brings remote collaborators into the local user's physical workspace as 3D avatars, enhancing social perception and helping distinguish comments from different authors. This spatial representation supports better context awareness, especially when reviewing conflicting suggestions. Unlike traditional tools that use cursors and names, our system displays remote users as avatars interacting with MR cues to improve immersion and social presence. [Cho et al. 2025, 2023].

Avatars include visual behaviors such as facing the local user, turning toward the collaborator's cursor, and showing comments in speech bubbles. By default, avatars appear near the monitor and face the user (Figure 1, Remote collaborator's 3D avatar). When a remote user is typing, the avatar turns towards the cursor's position with animation. The cursor position is tracked by the Text Asset Management module and transmitted over the network to maintain spatial alignment in MR. When an avatar is active, the associated 3D comment boxes are displayed as speech bubbles near the avatar in 3D space (Figure 1, Social interaction with avatars), reinforcing the collaborator's presence and linking their comments to their visual identity.

3 Conclusion

We presented MoreOverleaf, an MR collaborative writing tool with three key features: (1) *3D comment boxes* that spatially anchor annotations; (2) *Revision player* for visualizing document history; and (3) *MR avatars* that enhance social presence and authorship awareness. This approach helps users review editing spatially and temporally, clarifying each collaborator's contributions without cluttering the document. The revision player timeline slider supports understanding document history, while avatar-based cues improve author identification and perspective tracking. While promising, our system requires further validation through formal user studies. Future work will explore its impact on collaboration, focusing on social presence, awareness, and usability. This research contributes to HCI and 3D visualization by proposing a novel AR-driven solution for collaborative writing.

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